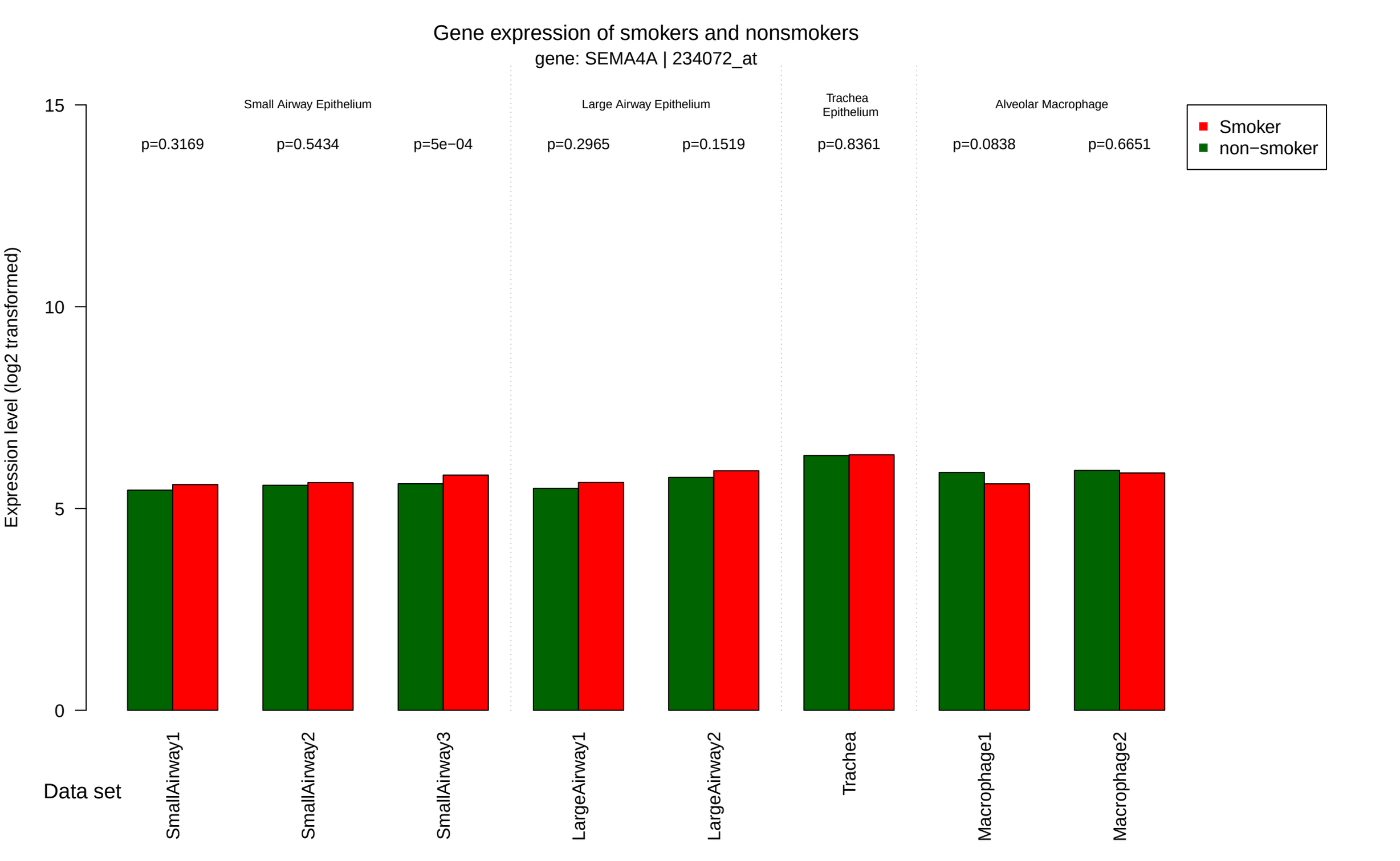
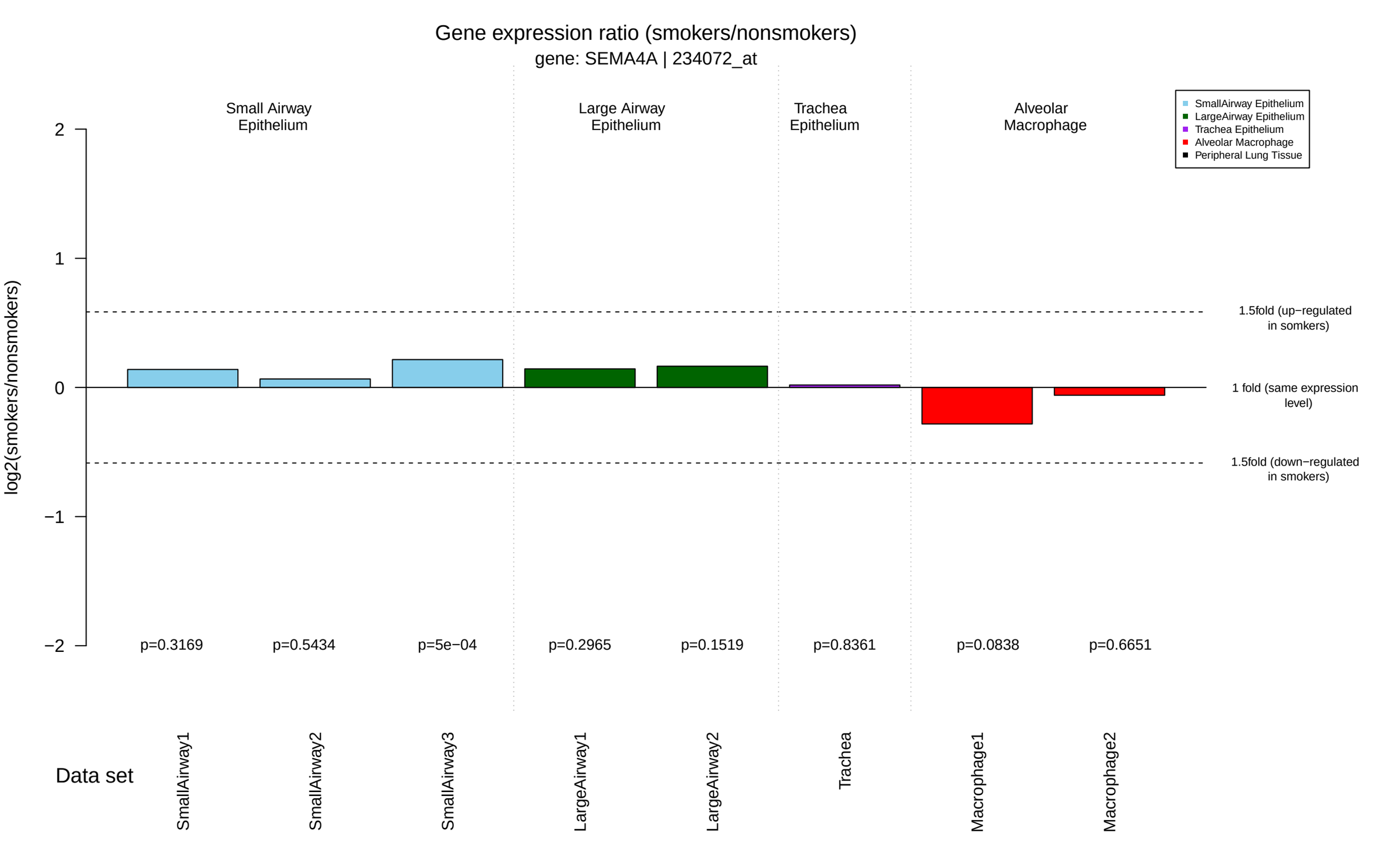
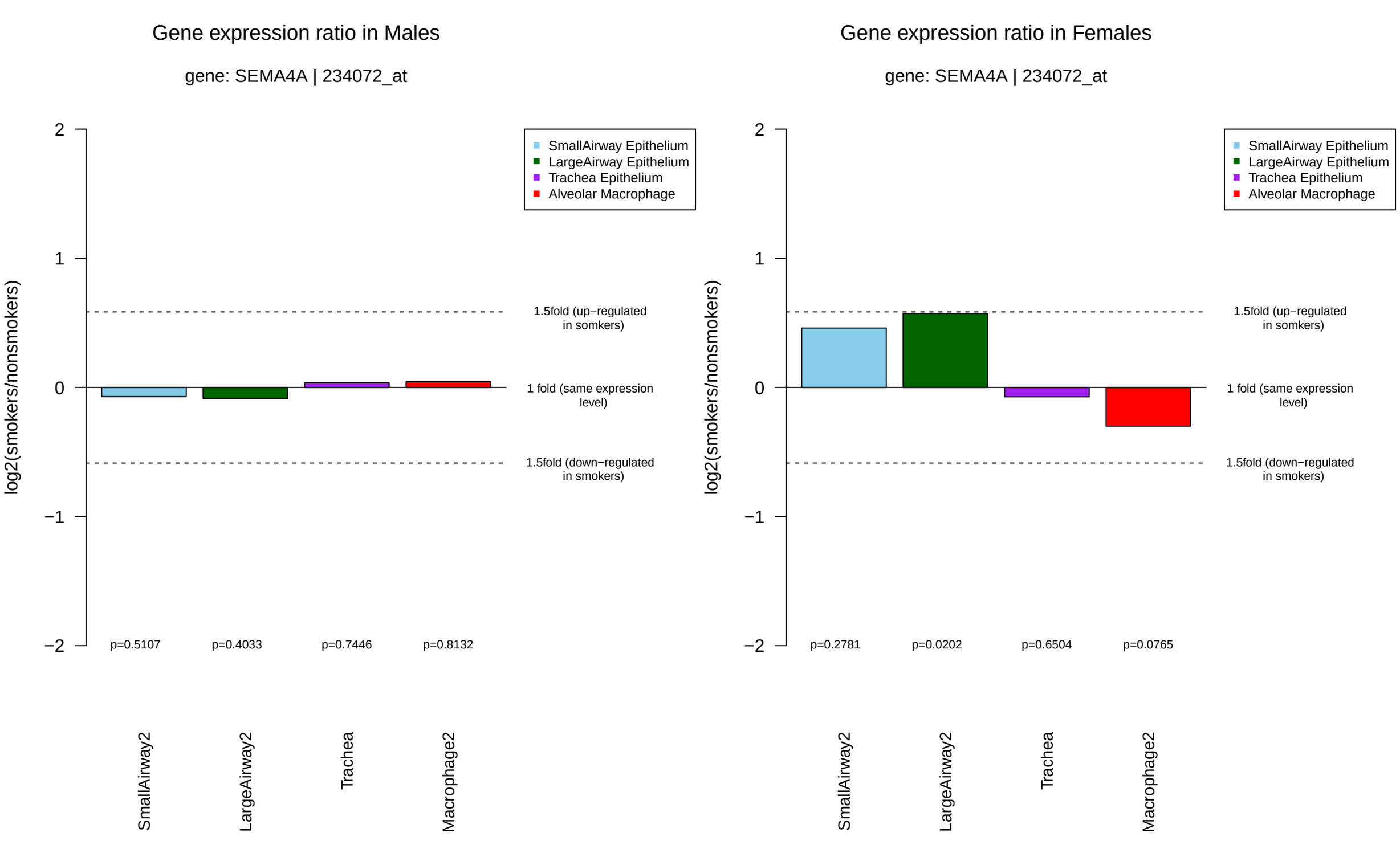
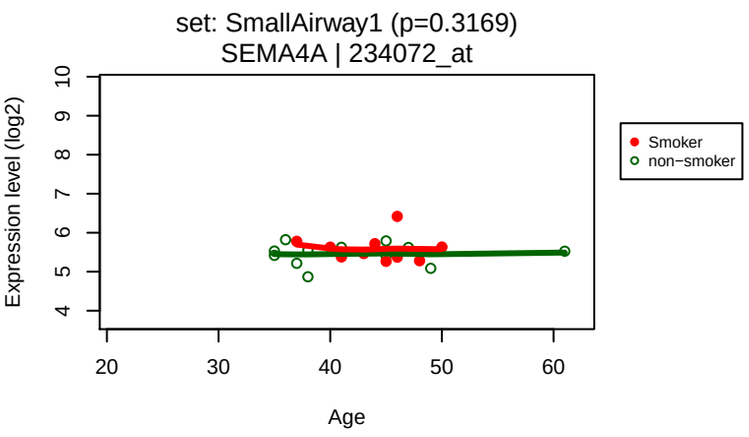
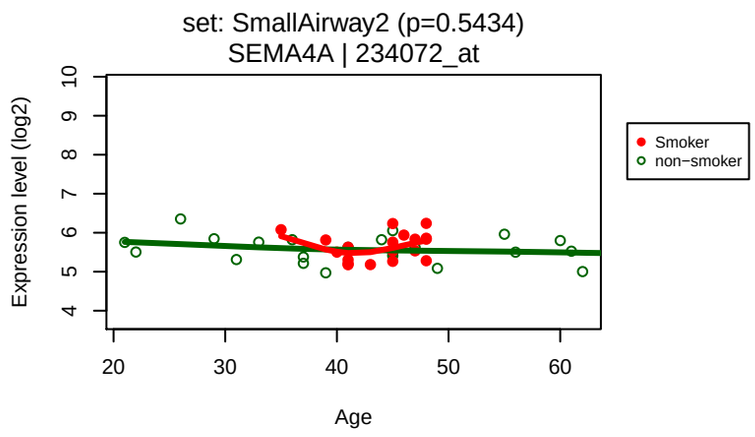
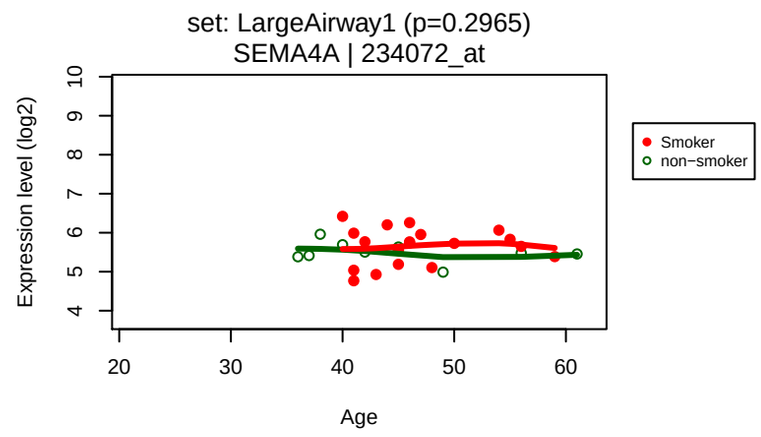

Gene expression of smokers and nonsmokers
gene: SEMA4A | 234072_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3169
p=0.5434
p=5e−04
p=0.2965
p=0.1519
p=0.8361
p=0.0838
p=0.6651
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

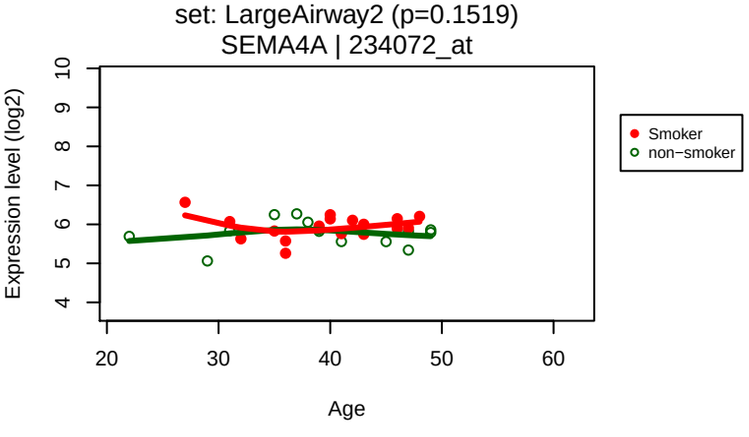

Gene expression ratio (smokers/nonsmokers)
gene: SEMA4A | 234072_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3169
p=0.5434
p=5e−04
p=0.2965
p=0.1519
p=0.8361
p=0.0838
p=0.6651
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

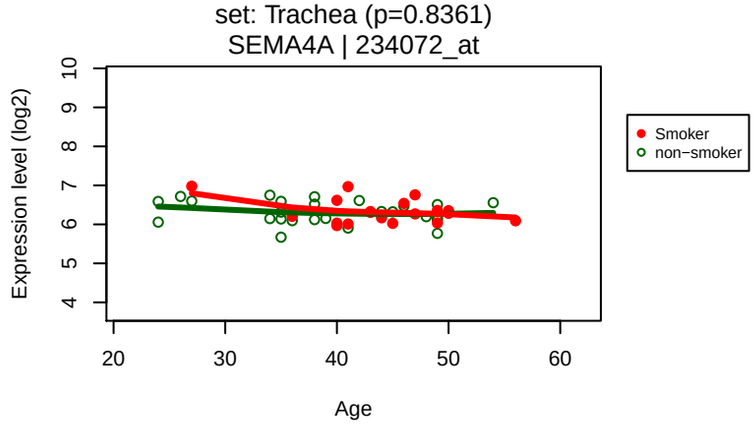

Gene expression ratio in Males
gene: SEMA4A | 234072_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5107
p=0.4033
p=0.7446
p=0.8132
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SEMA4A | 234072_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2781
p=0.0202
p=0.6504
p=0.0765
SmallAirway2
LargeAirway2
Trachea
Macrophage2

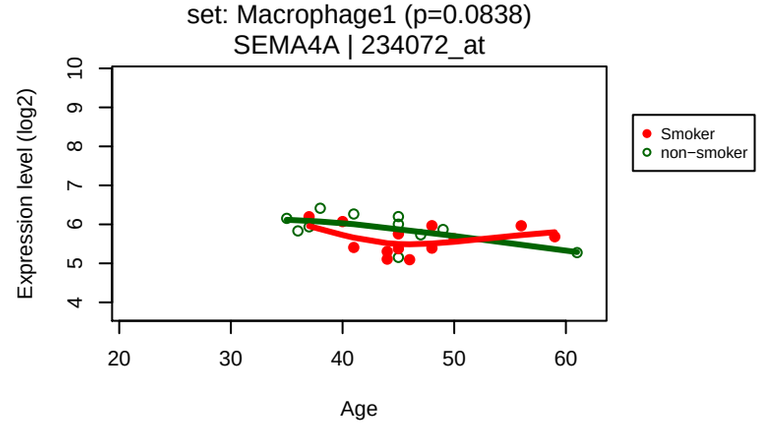

set: SmallAirway1 (p=0.3169)
SEMA4A | 234072_at
Expression level (log2)
Age
Smoker
non−smoker

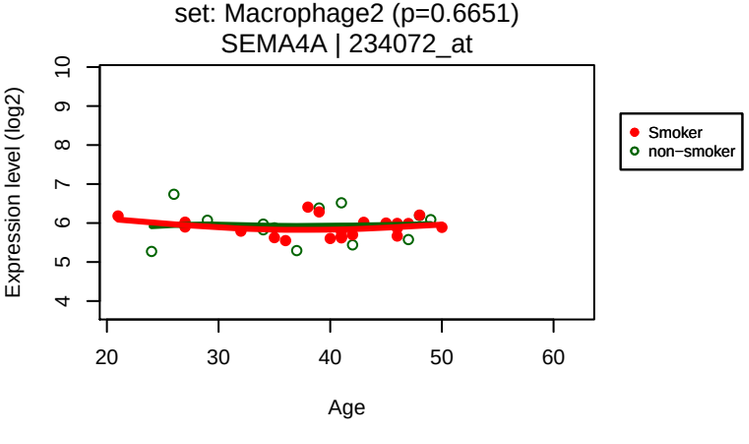

set: SmallAirway2 (p=0.5434)
SEMA4A | 234072_at
Expression level (log2)
Age
Smoker
non−smoker

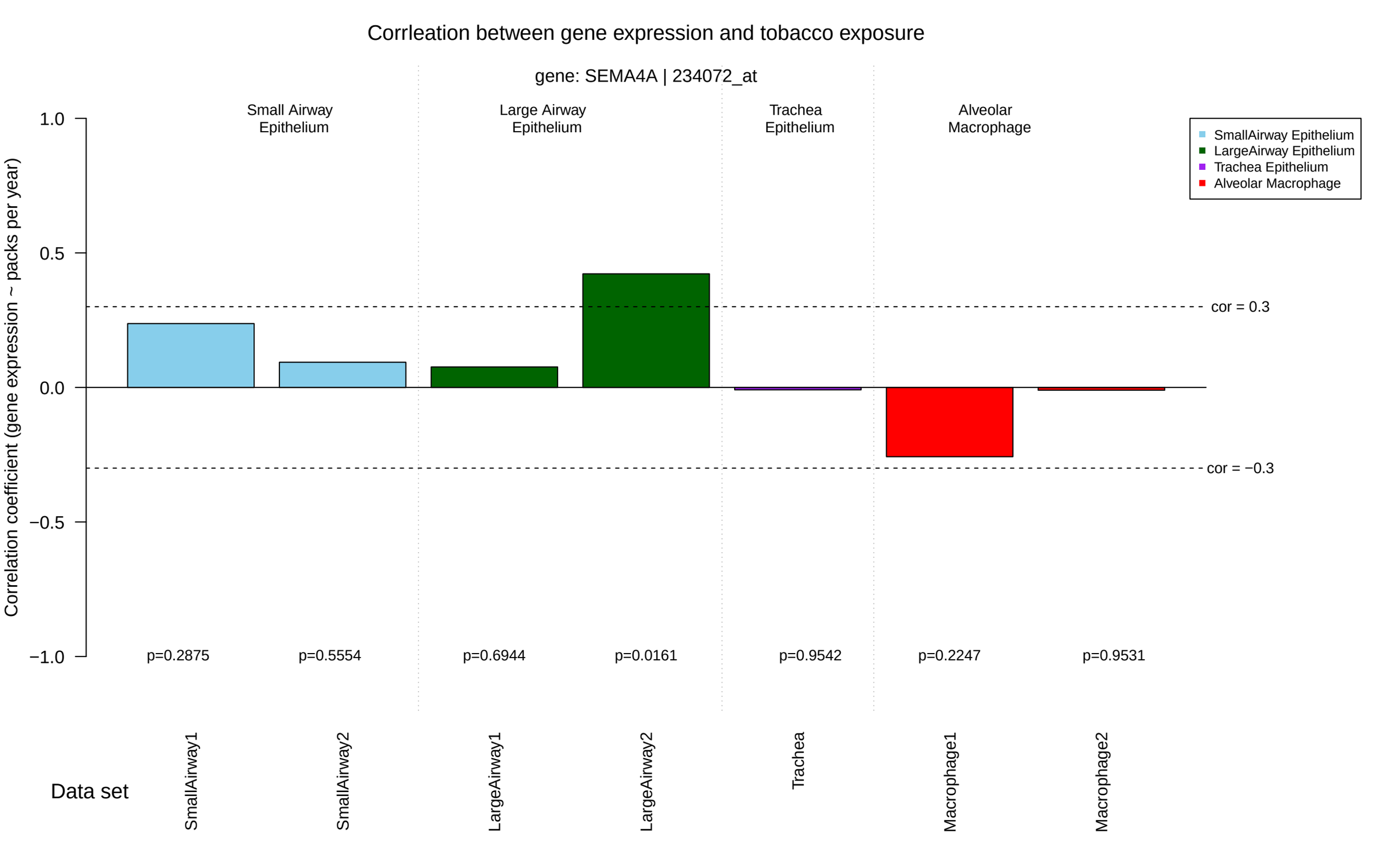

set: LargeAirway1 (p=0.2965)
SEMA4A | 234072_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1519)
SEMA4A | 234072_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8361)
SEMA4A | 234072_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0838)
SEMA4A | 234072_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6651)
SEMA4A | 234072_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SEMA4A | 234072_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2875
p=0.5554
p=0.6944
p=0.0161
p=0.9542
p=0.2247
p=0.9531
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2